## Alkane

1. Which represents an alkane
(a) $C_5H_8$ (b) $C_8H_6$
(c) $C_9H_{10}$ (d) $C_7H_{16}$

2. The decreasing order of boiling points is
(a) *n*-Pentane > iso-Pentane > neo-Pentane
(b) iso-Pentane > *n*-Pentane > neo-Pentane
(c) neo-Pentane > iso-Pentane > *n*-Pentane
(d) *n*-Pentane > *n*eo-Pentane > iso-Pentane

3. To prepare a pure sample of n-hexane using sodium metal as one reactant, the other reactant will be
(a) *n*-propyl bromide
(b) Ethyl bromide and *n*-butyl bromide
(c) Ethyl chloride and *n*-butyl chloride
(d) Methyl bromide and *n* -pentyl chloride

4. In the preparation of Grignard reagent from haloalkane, the metal
(a) *Mg* (b) *Zn*
(c) *Li* (d) *K*

5. Sodium acetate can be converted to ethane by
(a) Heating with $LiAlH_4$
(b) Electrolysing its aqueous solution
(c) Heating with sodalime
(d) Heating with calcium acetate

6. Which of the following compounds is used in antiknock compositions to prevent the deposition of oxides of lead on spark plug, combustion chamber and exhaust pipe
(a) Glycerol
(b) Glycol
(c) 1, 2-dibromoethane
(d) Benzene

7. Which of petroleum corresponds to kerosene oil
(a) $C_{15} - C_{18}$ (b) $C_{10} - C_{12}$
(c) $C_5 - C_9$ (d) $C_1 - C_9$

8. In the reaction $CH_3 - Br + 2Na + Br - CH_3 \rightarrow$, the product called
(a) Wurtz reaction
(b) Aldol condensation
(c) Perkin's reaction

(d) Levit reaction

9. Iodoethane reacts with sodium in the presence of dry ether. The product is
(a) Pentane (b) Propane
(c) Butene (d) Butane

10. Which of the following is oxidised by $KMnO_4$
(a) Methane
(b) Pentane
(c) Isobutane
(d) Neopentane

11. Which of the following has maximum stability
(a) $\overset{+}{C}H_3$
(b) $CH_3-\underset{+}{\overset{H}{\overset{|}{C}}}-H$
(c) $CH_3-\underset{+}{\overset{H}{\overset{|}{C}}}-CH_3$
(d) $CH_3-\underset{+}{\overset{CH_3}{\overset{|}{C}}}-CH_3$

12. The most volatile compound is
(a) 2, 2-dimethyl propane
(b) 2-methyl butane
(c) Isobutane
(d) *n*-pentane

13. In Wurtz reaction, the reagent used is
(a) *Na*
(b) $Na/\text{liquid } NH_3$
(c) $Na/\text{dry ether}$
(d) $Na/\text{dry alcohol}$

14. Which of the following has highest octane number
(a) *n*-hexane
(b) *n*-heptane
(c) *n*-pentane
(d) 2,2, 4-trimethyl pentane

15. What is freon-12
(a) Pesticide
(b) Refrigerant
(c) Solvent
(d) Lubricant

16. The petrol having octane number 80 has
(a) 20% normal heptane + 80% iso-octane
(b) 80% normal heptane + 20% iso-octane
(c) 20% normal heptane + 80% normal octane
(d) 80% normal heptane + 20% normal octane

17. Which of the following reactions will not give propane
(a) $CH_3CH_2CH_2Cl \xrightarrow[H_2O]{Mg/\text{ether}}$
(b) $CH_3COCl \xrightarrow[H_2O]{CH_3MgX}$
(c) $CH_3CH=CH_2 \xrightarrow[CH_3COOH]{B_2H_6}$
(d) $CH_3\underset{OH}{\underset{|}{CH}}-CH_3 \xrightarrow{P/HI}$

18. The shape of methane molecule is
(a) Linear
(b) Trigonal planar
(c) Square planar
(d) Tetrahedral

19. Which of the following shows only one brominated compound

(a) Butene-2
(b) 2, 2-dimethylpropane
(c) Butyne-1
(d) Butanol-3

**20.** Kerosene is used as fuel because it is
(a) Less volatile
(b) More volatile
(c) Cheap
(d) Abundantly available

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